

Unit 5 - PM

MC - Questions
Answered on D2L

MC	Written	Total
/13	/19	/32

Knowledge

1. Solve for x on the interval $0 \leq x \leq 2\pi$. Give the **exact solution(s)** in radian measure, where possible. Otherwise, round to the nearest hundredth of a radian.

a) $2 = 4 \cos^2 x - 1$

b) $2 \csc^2 x - 5 \csc x + 2 = 0$

(2)

(3)

2. a) Sketch the graph of $g(x) = -4 \cos \left[\frac{2}{3}x + \frac{\pi}{6} \right] + 1$ for one cycle on the grid provided. (1.5)

(5)

b) Determine:

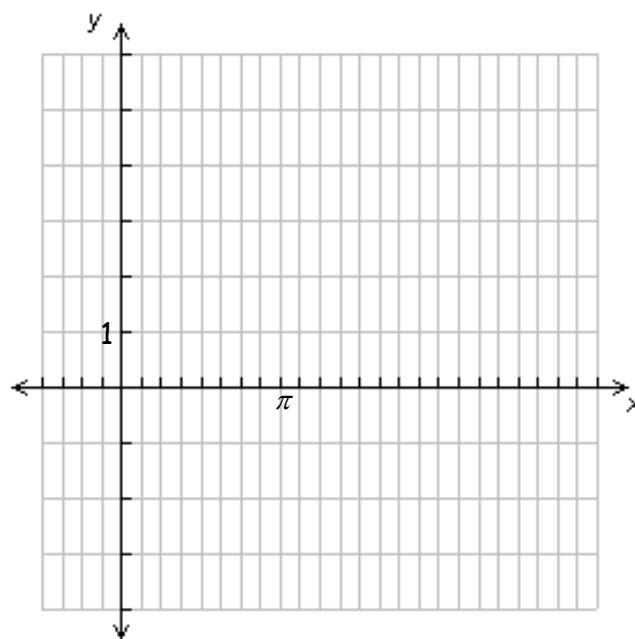
period: _____ (0.5)

phase shift: _____ (0.5)

vertical translation: _____ (0.5)

range: _____ (0.5)

mapping: $(x, y) \rightarrow$ _____ (1)



y-int _____ (0.5)

APPLICATION

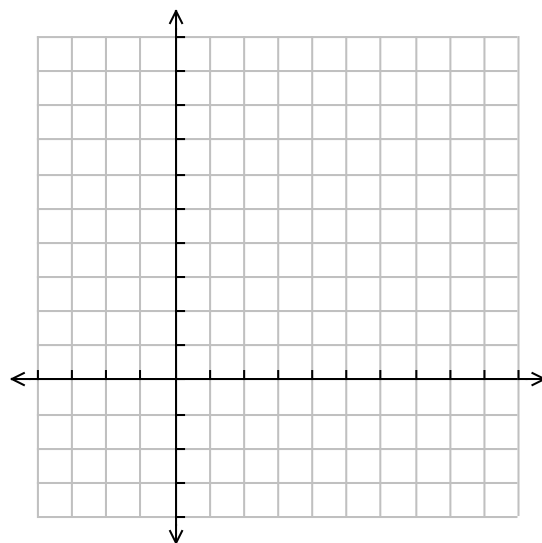
3. Maple Leaf Village used to have a large Ferris wheel that had a diameter of 40m. One revolution took 4min to complete.

- a. Sketch two cycles of a graph that represents the height of a rider above the ground, as a function of time, if the rider gets on at the bottom of the Ferris wheel at a height of 5 m at $t = 0$ min. Label your graph.

(2)

- b. Write an equation that describes this situation.

(2)



- c. Determine the height of the rider at 150 seconds.

(1)

THINKING

4. Solve for x on the interval $0 \leq x \leq 2\pi$. Give the **exact solution(s)** in radian measure if possible.

$$\sin\left(\frac{\pi}{4} + x\right) + \sin\left(\frac{\pi}{4} - x\right) = -1$$

(4)